3D–2D Medical Image Registration Using Deep Learning

Research Topic

Deep-learning-based registration of three-dimensional medical images with two-dimensional medical images using Deep Learning applications.

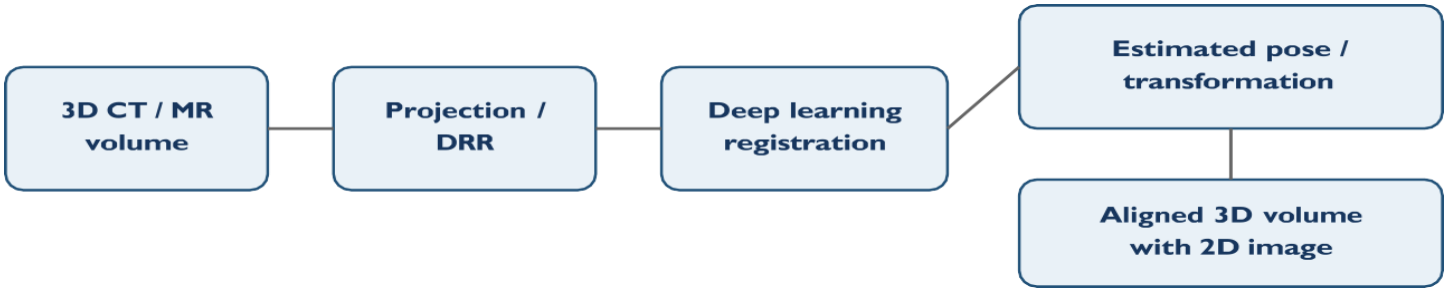
Student	Name	Registration Number
1	Ohil Raj S	261090700032
2	Rian Joy Mascarenhas	261090700052

1. Introduction

Medical image registration denotes the problem of estimating a spatial transformation that aligns anatomical information in different images. The fixed image is the reference image and the moving image is transformed to register corresponding anatomy. It is used in radiotherapy, motion analysis and surgical navigation. In the QIMS review, registration is divided into 3D-3D, 3D-2D and 2D-2D according to the spatial dimension, and 3D-2D registration is treated as a different problem from ordinary 3D volume registration. [1]

A 3D medical image such as a CT or MR volume contains volumetric information. An X-ray or fluoroscopic image is a 2D projection of anatomy. In 3D-2D registration, the problem is to estimate the 3D pose or transformation that a projection generated from the volume matches the observed 2D image. This is useful in cases where a preoperative 3D volume needs to be registered with an intraoperative 2D image, such as image-guided intervention or surgical navigation. [4,5]

Deep learning changes the computational formulation to learn from data image features, similarity functions, transformation parameters or iterative update steps. Initial work in CNN regression demonstrated direct estimation of 2D-3D transformation parameters, while later work investigated iterative learned refinement, self-supervised domain adaptation, differentiable rendering, anatomical feature reasoning and patient specific adaptation. [2,3,6–10]



2. Research Problem Statement

The central research problem is to establish an accurate spatial relationship between a 3D medical volume and a 2D medical image taking into account their different dimensions, projection ambiguity, and potentially differing appearances. The loss of depth information during the 2D image formation process renders the inverse projection from a 2D image to 3D pose highly ill-conditioned. [4,5]

Conventional intensity-based registration can involve iterative optimization of transformation parameters and can be sensitive to initialization, hyperparameters and the selected similarity objective. Learning-based approaches can reduce or replace parts of this iterative process, but their performance depends on training data, anatomy, acquisition geometry and the relationship between simulated and real images. Reviews of the field stress domain shift, generalization, confidence estimation and standardized evaluation as ongoing issues. [4,5]

Accordingly, the research problem is formulated as how a deep-learning-based 3D–2D registration approach could estimate accurate alignment efficiently and be reliable under different conditions in terms of appearance, anatomy, acquisition geometry, or data source presented during the training. Therefore, the project focuses on designing and validating the method rather than relying on the assumption that a particular architecture or dataset is optimal.

3. Literature Survey

The literature shows a progression from direct CNN regression toward hybrid and differentiable approaches that retain geometric knowledge while learning the difficult parts of registration. Miao et al. introduced hierarchical CNN regression using DRR and X-ray appearance to directly estimate transformation parameters, targeting slow optimization and limited capture range. [2]

Jaganathan et al. proposed an end-to-end deep iterative framework in which the update step is learned and combined with a point-to-plane correspondence solver. This preserves an iterative refinement structure while reducing dependence on a separate classical refinement stage. [3] Wei et al. addressed a different 3D–2D setting—2D ultrasound to 3D CT/MR for liver tumor ablation—using orientation classification followed by segmentation-based registration, illustrating that the learning formulation must reflect modality and clinical geometry. [4]

The systematic review by Unberath et al. is especially relevant to 3D–2D work because it identifies contextualization, representation learning, similarity modeling, direct parameter regression and verification as recurring learning themes. It also reports that many studies relied on synthetic DRRs and highlights the difficulty of preserving performance under domain shift. [5]

Van Houtte et al. demonstrated an end-to-end network that warps a 3D atlas so that its forward projections match biplanar radiographs, explicitly incorporating projective geometry. Jaganathan et al. later combined simulated training with feature- and pixel-space domain adaptation to reduce the synthetic-to-real gap for X-ray/CT registration. [6,7]

4. Literature Review Table

S.No.	Paper / Study	Authors & Year	Modality / Registration	Deep Learning Method	Dataset / Data	Main Finding	Limitation / Relevance
1	A CNN Regression Approach for Real-Time 2D/3D Registration	Miao, Wang & Liao (2016)	X-ray + CT/DRR; rigid 2D–3D	Hierarchical CNN regression; pose-indexed features	Clinical application datasets; DRRs and X-rays	Direct transformation prediction improved speed and capture range over intensity-based optimization.	Training is tied to defined pose zones and data distribution; foundational direct-regression baseline.
2	Deep Iterative 2D/3D Registration	Jaganathan et al. (2021)	X-ray/CT; rigid 2D–3D	Learned iterative residual refinement + point-to-plane solver	Simulated X-ray/CT registration data	Reported ~8 s runtime, 0.60±0.40 mm mean reprojection error and 97% success in the study.	Still depends on learned data distribution; iterative structure remains.
3	A deep learning approach for 2D ultrasound and 3D CT/MR image registration in liver tumor ablation	Wei et al. (2021)	2D US to 3D CT/MR; slice-to-volume	Orientation classification + segmentation; distance-map Dice loss	Liver tumor ablation imaging data	Shows that modality-specific learning can formulate difficult 2D-to-3D alignment without treating all modalities identically.	Different projection geometry from X-ray/CT; indirect relevance.
4	The Impact of Machine Learning on 2D/3D Registration for Image-Guided Interventions	Unberath et al. (2021)	Review of X-ray/CT and related 2D–3D tasks	Contextualization, representation learning, similarity modeling, direct regression, verification	48 studies in systematic review	Identified major learning themes and persistent issues including domain shift, limited real data and evaluation/reproducibility.	Review rather than a single registration model; directly informs gap formulation.
5	Deep Learning-Based 2D/3D Registration of an Atlas to Biplanar X-Ray Images	Van Houtte et al. (2022)	Biplanar X-ray + 3D atlas; 2D–3D	End-to-end dense deformation with projective/ inverse projective spatial transforms	Simulated DRRs from patient CT images	Reported high overlap and surface accuracy and handled non-orthogonal projection geometry.	Requires calibrated radiographs; evaluated primarily with simulated DRRs.
6	Self-Supervised 2D/3D Registration for X-Ray to CT Image Fusion	Jaganathan et al. (2023)	Real X-ray + CT; rigid 2D–3D	Self-supervised feature/pixel domain adaptation	Simulated CT projections + real X-ray images	1.83±1.16 mm accuracy and 90.1% success on real X-rays; addressed synthetic-to-real domain gap.	Performance remains dependent on adaptation setting and clinical data characteristics.
7	A Fully Differentiable Framework for 2D/3D Registration and the Projective Spatial Transformers	Gao et al. (2024)	Fluoroscopy/X-ray + CT; rigid 2D–3D	Projective Spatial Transformer; double-backward gradient-driven loss	Pelvis; simulation and real data	Extended capture range of conventional intensity-based registration; combined learned similarity with optimization.	Still uses a learned similarity stage followed by classical optimization; anatomy-specific validation.
8	Transformer-based 2D/3D Medical Image Registration for X-ray to CT via Anatomical Features	Qu et al. (2024)	Spinal X-ray + CT; affine 2D–3D	Feature-point detection + GNN + Swin Transformer	TCIA-derived data	Reported mean TRE up to 0.31 mm and ~0.6 s average runtime in the study.	Feasibility study focused on spinal anatomy and a defined acquisition setting.
9	A Review of Deep Learning-Based Three-Dimensional Medical Image Registration Methods	Xiao et al. (2021)	Broad 3D registration review; includes 3D–2D classification	Deep iterative, supervised and unsupervised DL taxonomy	68 DL-based 3D registration studies	Organizes DL registration by training regime and summarizes evaluation metrics and challenges.	Broader than 3D–2D; used here as technical foundation, not as the project itself.
10	Rapid Patient-Specific Neural Networks for X-ray to Volume Registration	Gopalakrishnan et al. (2026)	Real fluoroscopy/X-ray + CT/MR; rigid 2D–3D	Patient-specific neural network + differentiable physics-based rendering	Whole-body scans; real fluoroscopy across anatomy/sites	Foundation model plus ~5 min patient-specific adaptation; high accuracy reported across diverse clinical settings.	Recent work substantially reduces prior gaps; further independent validation and standardization remain important.

5. Research Gap

Even though deep-learning methods have demonstrated promising outcomes in 3D-to-2D medical image registration, it is still a challenge to achieve reliable and accurate registration under different imaging conditions. Current methods usually depend on synthetic digitally reconstructed radiographs (DRRs), fixed acquisition parameters, or anatomy-specific datasets, and this could hinder their ability to be applied to real intra-operative X-ray images. The discrepancies between the pre-operative 3D medical images and the intra-operative 2D X-rays also make cross-modal alignment more difficult. Although some recent approaches have looked into self-supervised learning and domain adaptation in an attempt to overcome these limitations, more research is required regarding the accuracy, computational efficiency, and robustness of the registration when the acquisition conditions vary. For this reason, the present study will examine a deep-learning-based rigid 3D–2D registration method for aligning pre-operative MRI data with intra-operative X-ray images, focusing on the evaluation of registration accuracy, inference time, and robustness in the face of realistic variations in imaging conditions. [5–10]

6. Objectives

1. To analyze existing deep-learning-based approaches for 3D-2D medical image registration, with emphasis on transformation-estimation strategies, synthetic-to-real domain limitations, and their applicability to rigid MRI to X-ray.

2. Investigate suitable representations of 3D volumes and 2D projections, including DRRs and projection-aware features, for learning-based registration.

3. Design or implement a deep-learning-based registration pipeline capable of estimating a clinically meaningful rigid transformation between a 3D volume and a 2D image.

4. Evaluate robustness across controlled changes in pose, image appearance and data source, and identify conditions under which registration performance degrades.

7. Block Diagram

